MILITARY SPECIFICATION

CHLORDIAZEPOXIDE HYDROCHLORIDE AND CLIDINIUM BROMIDE CAPSULES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Chlordiazepoxide Hydrochloride and Clidinium Bromide Capsules (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals (See 6.3).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be os use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 1800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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STANDARDS

FEDERAL

FED-STD-285 - Capsules (for Medicinal Purposes).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA).

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Applications for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

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3. REQUIREMENTS

3.1 Material. Shall be Chlordiazepoxide Hydrochloride and Clidinium Bromide Capsules, and shall be in accordance with all applicable requirements of FED-STD-285, and as specified herein.

S2. Classification. Shall be type I or type II.

S5.4 The following is added to this paragraph:

Shall be capsules containing 5.0 mg of chlordiazepoxide hydrochloride and 2.5 mg of clidinium bromide per capsule, *within the designed assay limits for the capsules.*

3.1.1 Assay for Chlordiazepoxide Hydrochloride and Clidinium Bromide. Shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amount of chlordiazepoxide hydrochloride and clidinium bromide when assayed as specified in 4.4.1.1 and 4.4.1.2 respectively.

3.1.2 Identification.

3.1.2.1 Chlordiazepoxide hydrochloride. A reddish color shall develop when contents of the capsule are tested as specified in 4.4.2.1.

3.1.2.2 Clidinium bromide. A light violet color shall develop when contents of capsule are tested as specified in 4.4.2.2.

3.1.3 Limit tests of clidinium bromide.

3.1.3.1 3-quinuclidinyl benzilate. The filled capsule shall contain no more than 0.75 mcg of 3-quinuclidinyl benzilate corresponding to 0.03 *percent of the clidinium bromide content* when determined as specified in 4.4.3.1.

3.1.3.2 3-hydroxy-1-methyl-quinuclidinium bromide. The filled capsule shall contain no more than 12.5 mcg of 3-hydroxy-1-methyl-quinuclidinium bromide corresponding to 0.50 percent of the clidinium bromide content when determined as specified in 4.4.3.2.

3.2 Ingredients.

3.2.1 Chlordiazepoxide hydrochloride. The chlordiazepoxide hydrochloride used in the manufacture of the capsules shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

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3.2.2 Clidinium Bromide. The clidinium bromide used in the manufacture of the capsules shall comply with the tests, standards, and requirements of NF XIII, which was official from 1 September 1970 to 30 June 1975.

3.3 Other ingredients. All other ingredients used in the manufacture of the capsules shall comply with S5.2 of FED-STD-285.

3.4 Expiration dating period. The expiration dating period shall be 36 months.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.3 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

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4.4 Tests.

4.4.1 Assay.

4.4.1.2 Chlordiazepoxide hydrochloride.

Reagents.

Methanol: Reagent grade.

Modified 3A Alcohol: Carefully mix 2.8 ml of concentrated sulfuric acid with sufficient anhydrous 3A denatured alcohol to make 1000 ml.

Procedure.

(Caution: Use low-actinic glassware throughout).

Accurately weigh a portion of capsule fill, equivalent to about 20 mg of chlordiazepoxide hydrochloride, into a 100 ml volumetric flask. Dissolve the chlordiazepoxide hydrochloride by shaking for 5 minutes with 50 ml of methanol and dilute to volume with methanol. Filter, discarding the first 25 ml. Pipet 10 ml of the filtrate into a 100 ml volumetric flask and dilute to volume with acidified 3A alcohol. Determine the absorbances of sample, and a concomitant chlordiazepoxide hydrochloride USP Reference Standard, at a concentration of approximately 20 mcg per ml, at the point of maximum absorbance between 311 and 312 nm, versus acidified 3A alcohol, using a 1 cm cell in a suitable spectrophotometer.

Calculation:

$$\frac{Au}{As} \times Cs \times \frac{100}{\text{mg of sample}} \times \frac{100}{10} \times \frac{\text{average weight per capsule (mg)}}{1} = \text{mg of chlordiazepoxide hydrochloride per capsule}$$

where: Au = absorbance of sample
As = absorbance of standard
Cs = final concentration of std.

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4.4.1.3 Clidinium bromide.

Reagents.

Thymol Blue: Dissolve 100 mg of thymolsulfonphthalein sodium salt in 50 ml of purified water and dilute to 100 ml with pH 8.8 buffer. Alternatively, triturate 100 mg of thymolsulfonphthalein with 2.15 ml of 0.1N sodium hydroxide. Dilute to 100 ml with purified water and adjust the pH to exactly 8.8.

pH 8.8 Buffer: (0.5M disodium phosphate): Dissolve 26.8 grams of disodium phosphate ($Na_2HPO_4.7H_2O$) in purified water and dilute to 200 ml. Check the pH and adjust, if necessary.

Chloroform: Reagent grade.

Standard Solution.

Weigh accurately about 50 mg of Clidinium Bromide Reference Standard sample into a 500 ml volumetric flask. Dissolve in and dilute to volume with purified water.

Sample Solution.

Weigh accurately a portion of capsule fill, equivalent to 10 mg of clidinium bromide, into a 100 ml volumetric flask. Add 50 ml of purified water and shake for 5 minutes. Dilute to volume with purified water. Filter through Whatman No. 2 filter paper, discarding the first 15 ml.

Procedure.

Into 125 ml separators proceed as follows:

	Blank	Standard	Sample
Standard Solution	---	10 ml	----
Sample Solution	---	----	10 ml
Purified Water	10 ml	----	----
pH 8.8 Buffer	10 ml	10 ml	10 ml
Thymol Blue Reagent	5 ml	5 ml	5 ml

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Extract each of the solutions in the separators with 3 x 25 ml portions and 1 x 20 ml portion of chloroform, shaking for 1 minute each time. Collect the chloroform extracts in separate 100 ml volumetric flasks and dilute to volume with chloroform. Mix well and measure the absorbance of the respective solutions at 410 nm against the blank in a 1 cm cell, using a suitable spectrophotometer.

Calculation:

$$\frac{Au}{As} \times \frac{\text{mg standard}}{500} \times \frac{100}{\text{sample weight (mg)}} \times \frac{\text{average capsule weight (mg)}}{1} = \text{mg of clidinium bromide per capsule}$$

Where: Au = absorbance of sample
As = absorbance of standard

4.4.2 Identification.

4.4.2.1 Chlordiazepoxide hydrochloride. Empty the contents of one capsule into a test tube. Add 5 ml of concentrated hydrochloric acid, 10 ml of purified water and warm to effect hydrolysis. Add 2 ml of a 0.1 percent aqueous sodium nitrite solution, 1 ml of a 0.5 percent aqueous ammonium sulfamate solution and 1 ml of a 0.1 percent aqueous N (1-naphthyl) ethylenediamine dihydrochloride solution.

4.4.2.2 Clidinium bromide. Empty the contents of one capsule into a test tube and add 0.5 ml of concentrated sulfuric acid. A yellowish solution is produced on shaking, which will decolorize and finally turn a violet color.

4.4.3 Limit tests of Clidinium bromide.

4.4.3.1 3-Quinuclidinyl benzilate.

Empty six capsules into a 100 ml beaker, add 3 ml of 1N hydrochloric acid and swirl. Add 4 grams of Celite 545 (acid washed) and mix with a spatula (sample or mixture).

To a glass chromatographic column (approximately 2.5 cm in diameter x 30-40 cm in length) containing a glass wool plug, add a mixture of 2 grams of Celite 545 (acid washed) and 1 ml of 1N hydrochloric acid. Pack lightly with a plunger, then insert the sample mixture. Dry wash the 100 ml beaker with 0.5 - 1 gram of Celite 545 and place on top of the column. Pack lightly with a plunger and add some glass wool on top. Insert the narrow end of the column into a 125 ml separator. Elute the column with 100 ml of chloroform which was distilled over 1N sulfuric acid and subsequently saturated with purified water.

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Prepare a 5 percent aqueous solution of ascorbic acid in purified water and extract the chloroform in the separator with 20 ml of the ascorbic acid solution. Save the ascorbic acid extract. Again extract the chloroform layer with 15 ml of the 5 percent ascorbic acid solution. Combine the ascorbic acid extracts in the separator and discard the chloroform layer. Add carefully solid sodium bicarbonate to neutralize the acid. Check the pH with pH paper to be sure the solution is mildly alkaline.

Extract the alkaline solution with 2 x 25 ml of chloroform and pass the chloroform extracts through a dry fluted filter paper (about 10-cm diameter) into a 100 ml beaker. Evaporate the chloroform on a steam bath, using a stream of nitrogen, to dryness. Transfer the residue to a 1.0 ml graduated micro tube with a glass stopper, using methanol for the transfer and dilute to volume (Sample solution).

Thin-Layer Chromatographs. (Run in duplicate on two separate plates.)

Adsorbent:	Silica Gel G - approximately 250 microns in thickness.
Solvent:	Methanol
Standard Solution:	Dissolve 3 mg of 3-quinuclidinyl benzilate in 100 ml of methanol. Apply as separate spots to the starting line of the thin layer plate 0.01 ml [(0.3 mcg (0.02 percent)], 0.015 ml [(0.45 mcg) (0.03 percent)] and 0.02 ml [(0.6 mcg (0.04 percent)].
Sample Solution:	Apply 0.1 mg of the Sample Solution (equivalent to 1.5 mg of clidinium bromide).
Development:	Ascending for approximately 15 cm in a paper-lined saturated tank. Remove from the tank and air dry.
Detection:	Spray the plate with a solution of potassium iodoplatinate T.S. and allow the spots to develop for at least 10 minutes.

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Mobility:	Approximate R_f
3-quinuclidinyl benzilate	0.3-0.4
Chlordiazepoxide hydrochloride	0.8

If a spot appears in the sample at the R_f corresponding to 3-quinuclidinyl benzilate, the intensity and size of the spot shall be less than the 0.03 percent 3-quinuclidinyl benzilate standard spot (equivalent to less than 0.75 mcg of 3-quinuclidinyl benzilate per capsule).

4.4.3.2 3-hydroxy-1-methyl quinuclidinium bromide.

Thin-layer chromatography. (Run in duplicate on two separate plates.)

Adsorbent: Silica Gel G approximately 250 microns in thickness.

Solvent System: Acetone:methanol:purified water: concentrated hydrochloric acid (70:20:5:5) (v/v)

Development A: Condition a thin-layer plate by developing in the solvent system for 15 cm initially without any sample. Remove the plate from the chromatographic chamber and dry at 105°C. for 15 minutes.

Sample Solution: Empty 10 capsules into a glass-stoppered centrifuge tube and add 5.0 ml of a solution of anhydrous ethanol:cyclohexane (1:1) (v/v). Warm the tube to about 50°C. with shaking. Centrifuge the resulting mixture and transfer the supernatant liquid to another stoppered tube. Extract the remaining powder two more times in the same manner, combining all the extracts to dryness with the aid of a stream of nitrogen, while warming gently. Dissolve the residue in 0.5 ml of methanol (Sample Solution).

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Standard Solution:	Carefully weigh 12.5 mg of standard 3-hydroxy-1-methyl-quinuclidinium bromide and dissolve in 1.0 ml of methanol (3-hydroxy-1-methyl-quinuclidinium bromide Standard Solution). Dissolve two separate samples of 50 mg each of standard clidinium bromide in 1.0 ml of methanol (Solutions I and II, respectively).
Standard Solution I (0.25 percent 3-hydroxy-1-methyl-quinuclidinium bromide):	To the standard solution of clidinium bromide (I), add 0.01 ml (125 mcg) of the standard solution of 3-hydroxy-1-methyl-quinuclidinium bromide.
Standard Solution II (0.50 percent 3-hydroxy-1-methyl-quinuclidium bromide):	To the Standard Solution of clidinium bromide (II), add 0.02 ml (250 mcg of the standard solution of 3-hydroxy-1-methyl-quinuclidinium bromide.
Application of Solutions to the Thin-Layer Plate and Development B:	Cool the dried plate as obtained under Development A to room temperature and apply to the starting line of the thin-layer plate 0.04 ml of the sample solution and adjacent 0.04 ml of aliquots of standard solutions I and II. Then place the plate in an unsaturated chromatographic chamber and develop, ascending, for 10 cm. Remove from the tank and dry the plate at 105°C. for 10 minutes.
Detection:	
3-hydroxy-1-methyl-quinuclidinium bromide:	Cool the plate to room temperature and spray the chromatograph with a solution of Dragendorff reagent (sulfuric acid modified).1/ Dark spots shall appear within 5 minutes.
1/ Dragendorff Reagent (sulfuric acid modified):	Solution A: Dissolve 0.85 grams of bismuth nitrate in a mixture of 10 ml of glacial acetic acid and 40 ml of purified water. Solution B: Dissolve 8.0 grams of potassium iodide in 20 ml of purified water. Spray Solution: Mix 5 ml of Solution A and 5 ml of Solution B and dilute to 50 ml with 10 percent sulfuric acid.

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Mobility:	
3-hydroxy-1-methyl-quinuclidinium bromide:	0.4
Clidinium Bromide:	0.5

If a spot appears in the sample solution chromatogram at the R_f corresponding to 3-hydroxy-1-methyl-quinuclidinium bromide in the standard solution chromatogram, the intensity and size of the spot shall be less than 0.5 percent with respect to 3-hydroxy-1-methyl-quinuclidinium bromide (equivalent to less than 12.5 mcg of 3-hydroxy-1-methyl-quinuclidinium bromide per capsule).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 (See 6.3) and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (See 6.1).

5.2.1 Unit of issue. One bottle containing 500 capsules, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatability that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit packaging. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate packaging. Twelve (12) units of issue shall be packaged as specified in 5.2.8 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.3 Packing. Packing shall be level A, B or C, as specified (See 6.1).

5.3.1 Level A. Ninety-six (96) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Ninety-six (96) units of issue shall be packed as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

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5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. In addition, Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15°-30°C (59°-86°F)".

5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15°-30°C (59°-86°F)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) National Stock Number.

(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-00-074-4692	CHLORDIAZEPOXIDE HYDROCHLORIDE AND CLIDINIUM BROMIDE CAPSULES, 500s

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1665

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-37501 CHLORDIAZEPOXIDE HYDROCHLORIDE AND CLIDINIUM BROMIDE CAPSULES

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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